

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2025-26 (NEP)

Computer Science**Paper Code: CSCMV-31 : IT Tools and Applications**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Answer all questions as directed. Marks are indicated against each question.***Question 1. Short Answer Questions:** [14]

- (a) What do you mean by Entity integrity constraint? [2]
- (b) What is the use of OLE Object data type in MS Access? [2]
- (c) What is the role of Input Masks in MS Access? [2]
- (d) What is the role of Animation Pane in PowerPoint? [2]
- (e) What is a Placeholder in PowerPoint? [2]
- (f) What is the purpose of data validation in Excel? [2]
- (g) What is YTD metric in sales data? Name a chart which can be used to represent hierarchical data. [2]

Question 2. Word Processing & Excel Short Concepts: [14]

- (a) What is absolute reference type in Excel? What is the output of the following formula:
=IF(SUM(12, 8, 10)>30, "Above-" & ROUND(MIN(9, 4, 6)*1.5,1), IF(AVERAGE(4, 8)<6, "Below-" & MAX(2, 3, 1), "Neutral"))? [2]
- (b) How stacked column chart is different from 100% stacked column chart? What is sparkline in Excel? [2]
- (c) Explain the use of the “Format Painter” tool in MS Word. [2]
- (d) What is the “Thesaurus” tool, and how does it assist in writing in MS Word? [2]
- (e) Explain how to create a multi-column layout (like a newspaper) in MS Word. [2]
- (f) Explain how the “Replace All” feature works and identify a potential risk of using it in MS Word. [2]
- (g) How do you change the case of selected text (e.g., from lowercase to UPPERCASE) in MS Word? How can you select non-adjacent pieces of text simultaneously? [2]

Question 3. Fundamentals of Word Processing: [14]

- (a) Define word processing and list three primary advantages over manual typing. How a Word Processor is different from a simple Text Editor? [4]
 - (b) What are “Styles,” and how do they help in maintaining document consistency? Explain how to modify an existing style. [3]
 - (c) What are “Indents,” and how do they differ from “Margins”? Describe how to set custom page margins for a formal report. [3]
 - (d) Explain the purpose of “Headers” and “Footers.” [2]
 - (e) Explain the application of word processing in the field of Education. [2]
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Question 4. Advanced Document Formatting & Mail Merge: [13]

- (a) What is a “Document Template”? How does a template differ from a regular document file? Explain the steps you need to follow to create your own Document Template. [3]
 - (b) What is a “Page Break,” and how is it different from a “Section Break”? Why would a user need to use different section breaks in a single document? [3]
 - (c) Define “Mail Merge” and explain its primary purpose. What are the three main components of a Mail Merge process? What is a “Data Source”, and what types of files can be used as one? [5]
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Question 5. Business Communication & Excel PivotTables: [17]

- (a) Discuss the role of word processing in Business Communication (reports, memos, letters). [3]
 - (b) Given an Excel dataset with columns: `CustomerName`, `ShippingCost`, `OrderDate`, `DeliveryDate`, `Region`, `Sales`, `PaymentMethod`, `PromotionCode`:
What is PivotTable? Mention any 2 benefits of PivotTable. [4]
 - (c) What kind of data is filled in columns, rows, values, and filter sections in PivotTable? [4]
 - (d) From the given dataset, give 2 separate sets of examples of data fields that can be filled in columns, rows, values, and filter sections in PivotTable. [4]
 - (e) What is calculated field in PivotTable? Prepare a calculated field from the given dataset. [2]
 - (f) What is the difference between filter and slicer for a PivotTable? [2]
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Question 6. Excel Data Analysis in School Management: [14]

Context: A school keeps an Excel workbook to track student marks, attendance, and grades. Columns include Student ID, Student Name, Subject, Marks Obtained, Total Marks, Attendance Percentage, and Grade.

- (a) Explain how date and time functions like `TODAY()`, `NOW()`, and `NETWORKDAYS()` can help track attendance and submission deadlines. [3]
 - (b) How can text functions like `LEFT`, `MID`, `UPPER`, `LOWER`, `PROPER` and `LEN` be used to process student IDs or names? Give an example for each. [3]
 - (c) What is the use of Text to Columns? Give an example with student data. [2]
 - (d) Explain Flash Fill. Give an example of how a teacher could use it to split full student names into first and last names. [2]
 - (e) Explain the difference between `COUNT`, `COUNTA`, `COUNTBLANK` and `COUNTIFS` along with example with respect to student data. [4]
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Question 7. Presentation Software (PowerPoint): [12]

- (a) What do you mean by Animation in PowerPoint? Describe Entry, Exit, and Emphasis animations with suitable examples. [5]
 - (b) Explain Rehearse Timings in PowerPoint? How does Rehearse Timings help in presentation delivery? [4]
 - (c) Explain the purpose and advantages of Slide Sorter View in PowerPoint. [3]
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Question 8. Database Management Systems (MS Access): [12]

- (a) Explain primary key and foreign key with suitable example. [5]
 - (b) Explain validation rules and validation text in MS Access with suitable examples. [4]
 - (c) Explain the role of tables, queries, forms, and reports in database management. [3]
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